

3.1.1 Particles and Radiation

- **Constituents of the atom**

Proton, neutron, electron.

Their charge and mass in SI units and relative units. Specific charge of nuclei and of ions. Atomic mass unit is not required.

Proton number Z , nucleon number A , nuclide notation, isotopes

- **Stable and unstable nuclei**

The strong nuclear force; its role in keeping the nucleus stable; short-range attraction to about 3 fm, very-short range repulsion below about 0.5 fm.

Equations for alpha decay and β^- decay including the antineutrino.

- **Particles, antiparticles and photons**

Candidates should know that for every type of particle, there is a corresponding antiparticle. They should know that the positron, the antiproton, the antineutron and the antineutrino are the antiparticles of the electron, the proton, the neutron and the neutrino, respectively.

Comparison of particle and antiparticle masses, charge and rest energy in MeV

Photon model of electromagnetic radiation, the Planck constant,

$$E = hf = \frac{hc}{\lambda}$$

Knowledge of annihilation and pair production processes and the respective energies involved. The use of $E = mc^2$ is not required in calculations.

- **Particle interactions**

Concept of exchange particles to explain forces between elementary particles.

The electromagnetic force; virtual photons as the exchange particle.

The weak interaction limited β^- , β^+ decay, electron capture and electron-proton collisions; W^+ and W^- as the exchange particles.

Simple Feynman diagrams to represent the above reactions or interactions in terms of particles going in and out and exchange particles.

- **Classification of particles**

Hadrons: baryons (proton, neutron) and antibaryons (antiproton and antineutron) and mesons (pion, kaon).

Hadrons are subject to the strong nuclear force.

Candidates should know that the proton is the only stable baryon into which other baryons eventually decay; in particular, the decay of the neutron should be known.

Leptons: electron, muon, neutrino (electron and muon types).

Leptons are subject to the weak interaction.

Candidates will be expected to know baryon numbers for the hadrons. Lepton numbers for the leptons will be given in the data booklet.

- **Quarks and antiquarks**

Up (u), down (d) and strange (s) quarks only.

Properties of quarks: charge, baryon number and strangeness.

Combinations of quarks and antiquarks required for baryons (proton and neutron only), antibaryons (antiproton and antineutron only) and mesons (pion and kaon) only.

Change of quark character in β^- and β^+ decay.

Application of the conservation laws for charge, baryon number, lepton number and strangeness to particle interactions. The necessary data will be provided in questions for particles outside those specified.